

Examining how data becomes information for an upcoming opponent in football

Saumya Mehta , Philip Furley , Dominik Raabe, and Daniel Memmert

International Journal of Sports Science & Coaching
2024, Vol. 19(3) 978–987
© The Author(s) 2023
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/17479541231187871
journals.sagepub.com/home/spo



Abstract

As the sport industry witnesses a surge in the type and volume of data-driven decisions, the general question of the process of information development remains: how is data used to develop meaningful information? And does the presence of novel quantitative data sources lend greater objectivity to match analysis? Study 1 examines how 12 football analysts use the same qualitative (video) and quantitative (event and position) data to develop information constituting a typical opponent report for an upcoming match, while Study 2 investigates the agreement between grade evaluations of these opponent reports by numerous professional coaches. Findings of Study 1 through independent-samples *t*-tests ($t(18) = 3.922$, $p = 0.001$) indicate a clear dominance of qualitative video data over quantitative event and position data in all opponent reports. Despite the presence of quantitative data sources, analysts tend to prefer annotated video data. Possible relations to previous experience and familiarity with data, coach–analyst preferences and biases are discussed. Results from Study 2 show extremely weak intra-class correlations (ICC) ($r = 0.147$; $p = 0.011$) between different grades awarded to the same video, depicting a clear lack of agreement in what coaches consider a good opponent report. Furthermore, coaches most valued the comprehensibility and relevance of the report. No significant associations were found between use of either data type and better grades. The subjectivity of the coaching process highlighting preferences regarding data validity and negotiations of adopting new key performance indicators (KPIs) is discussed, alongside limitations of the sample as well as the level of coach–analysts involved.

Keywords

Coach-analyst interaction, decision making, key performance indicators, soccer, tactical behaviour, video

Introduction

Performance analysts are now considered a mainstay of any coaching staff in football, not only at the highest level of sport but also trickling down into academy setups and youth development models. It was not long ago that this would have been difficult to find, but as the value and application of data gradually increased, clubs now have departments of analysts dedicated to deciphering everything possible about an opponent. This has led to a considerable increase in the volume as well as types of data, leading to the development of complex key performance indicators (KPIs). The adoption of these KPIs has even been associated with improved on-field performance.¹ While this has taken the sophistication and scope of match analysis to more complex levels, there is still a lack of clarity regarding the process, outcome and application of opponent performance analysis and its related pedagogical underpinnings.²

This is the case particularly in professional coach–analyst settings where valid expert notations are based on

trust and consist of a small, closed group.² Moreover, a large part of current match analysis research focuses on methods of analysis rather than the human factors involved in the decision-making or the complex nature of coaching as a whole.^{3,4} While the methods of analysis are critical in producing accurate or desired KPIs, the adoption of these data-driven insights, and the degree to which they are implemented, is usually based on preferences of

Reviewers: Jonathan Connor (James Cook University, Australia)
Denise Martin (Technological University Dublin, Republic of Ireland)

Institute of Exercise Training and Sport Informatics, German Sport University Cologne, Köln, Germany

Corresponding author:

Saumya Mehta, Institute of Exercise Training and Sport Informatics, German Sport University Cologne, Am Sportpark Mungersdorf 6, 50933 Köln, Germany.
Email: s.mehta@dshs-koeln.de

coaches and the pre-determined protocols of the staff.^{5,6} This is also why the performance analytical process has previously been referred to as a 'black box' owing to the complexity, secrecy and opaqueness of the coach-athlete decision-making system.⁷ More recently, a comprehensive analysis of the challenges within the process shows a multitude of factors that affect the applied performance analysis practice, such as general role ambiguity, skill set uncertainty and socio-cultural factors within the club setting.⁸⁻¹⁰

These socio-cultural factors, such as social hierarchies, belief systems, power networks and social capital associated with experience and reputation, all play a role in determining the development and dissemination of information within the entire group.⁸ This also relates to the idea of co-creation of value, where value is created through the exchange of resources and perspectives of several actors within a system and not through isolated efforts of individual actors.¹¹ Consequently, it is not surprising that the majority of analysts have to review all the data-driven information with coaches before disseminating the same to the players,¹² while the literature also supports the need for having clarity and shared knowledge of coaches' philosophy in making sure that feedback is precise and effective.¹³⁻¹⁵ Given recent work¹⁰ that sheds light on the co-creative and interactive nature of knowledge development in sport performance analysis, as well as the importance of the socio-cultural context in which it develops, a structured inquiry into the process of developing such knowledge from raw data or KPIs may be highly valuable.

While the use of new, quantitative KPIs may possibly lend greater objectivity to the insights drawn through them, there may not be any uniformity in the implementation of these insights amongst different coaches, as KPIs are often selected upon subjective preferences based on instinct or coaching philosophy.^{5,6} The traditional method of performance analysis involves the use of qualitative video data – notations of actions tagged onto video feeds that can then be aggregated to break the game into different phases and actions. Other data sources that utilize newer technology provide quantitative data on performance, such as data gathered from global positioning systems (GPS) or local positioning measurements (LPM) of players. They are used to summarize and describe individual performance through simple measures like frequency and intensity but also more complex KPIs such as expected goals (xG) and space control.^{16,17} As the sophistication of the KPIs, general analysis and tactical ideologies as well as the knowledge of the personnel involved keeps increasing,^{1,17} there is a need to understand whether this exponential growth in data types and volume relates to more objective forms of analysis. While insightful findings have been established through detailed interviews and process descriptions in the past,^{7,12} the same has not been examined in a coach-analyst setting while upholding

context familiarity by developing a typical opponent report in the build-up to a game.

To this end, with the purpose of mimicking a representative scenario in the data flow between analysts and coaches, we conducted a two-part study by first asking different analysts to prepare an opponent report from a homogenous dataset (Study 1) and subsequently analysing the reports for content overlap and having football coaches evaluate the reports (Study 2). Study 1 would help us understand how different analysts select and include (or do not include) different data types within their reports when provided with the same data. Study 2 would potentially explain what coaches look for within different opponent reports but also whether there is a similarity in how different coaches look at the same opponent report. The studies were developed in accordance with guidelines by Smith and McGannon^{18,19} for conducting rigorous qualitative research in sports. Given that the analysts and coaches were associated with different professional clubs, all of which have their own football philosophies, socio-cultural systems and protocols,⁸ it was expected that different analysts would show different data choices, while different coaches may also evaluate the same report differently. Furthermore, owing to the gap in the transfer of research knowledge into practice, a general dominance of video data use was expected,¹ while differing amounts of event and position data were expected to be used in the creation of the opponent reports.

In today's world of performance analysis, where data is exponentially increasing constantly, such research may be highly pertinent in terms of its application within sport backroom staff but may also provide a novel way to examine whether the inclusion of new and complex quantitative or objective data parameters lends greater objectivity to the match analysis process as a whole. Besides the application of this work in the domain of performance analysis in sports, the process of analysing data selection and development of information may be highly pertinent in understanding how capacity-limited humans operate in a world of exponentially growing data and how this relates to developing meaningful information, shedding light on concepts of human attention like preferences, biases and selective attention.²⁰ Moreover, examining the human factors within the process of turning data into information could provide a foundation for developing methods/paradigms of data-driven decision-making such as actuarial judgment, thereby addressing the human involvement in the creation of meaningful information with data.

Study 1: analysing the creation of typical opponent reports as preparation for an upcoming match

In our first study, we were mainly interested in the amount of uniformity of created opponent reports. As every analyst

received the exact same data, we could analyse the amount of subjectivity in using this data to create opponent reports. Given recent reports²¹ on people's subjective biases in interpreting different sources of data, we expected that the created opponent reports would not exhibit much uniformity in how the same data sources were used in creating opponent reports.

Method

Sample and data collection. The strategy and method of sampling were determined based on guidelines provided by Robinson²² about participant sampling in qualitative research. Twelve professional football analysts currently associated with professional football clubs in Germany were provided with a homogenous dataset. They were asked to develop an opponent report just like they would in their day-to-day work. Each participant was first sent an online link with a checklist of demographic information hosted by SoSci Survey (soscisurvey.de). Once complete, the participant was then provided access to a dataset describing several performance metrics of a professional team playing in the American Major League Soccer. The data included annotated video data that formulated the qualitative data, while the quantitative data was composed of event data and position data. Event data included metrics of each match played by the team that season in terms of discrete events within the game such as shots (attempted, on target and off target), passes, crosses, clearance, blocks, etc. Position data included timestamp-based records of player positions on-field with the help of GPS tracking data. The participants were informed that they had to develop an opponent report using Microsoft PowerPoint (Windows, 2019) as if they were facing the specific team whose data they had received. It could not exceed a total of 15 minutes in length and could include all or none of the aforementioned metrics. The participants were expected to make selections of metrics based on their usual working method and protocol, which would be followed every week by their team. Before the start of the study, all the participants were also asked to provide consent regarding the data and information that they would provide as part of the study, as well as not sharing the provided dataset.

Data analysis. The qualifying opponent reports (hereby referred to as "QR1 to QR12") created by the analysts were examined based on the time spent by the analyst on different types of data as well as different domain-specific themes that were developed after reviewing the range of information that was included in the report. Time spent on different data sources was determined by the time in seconds (and percentage of total time) each participant spent speaking about the data type or using a KPI associated with that data type during the presentation. The data sources

were broken down into qualitative data that consists of video data and quantitative data that consists of event data and position data. An independent-samples *t*-test was used to assess whether there is a significant difference in the time spent on qualitative and quantitative data in all the included opponent reports. Furthermore, the information provided in all the reports could largely be broken down into the following themes: general information (line-ups, formations and player information), opponent tactical behaviour (in defensive, offensive and transition phases), recommendations for own team (in defensive, offensive and transition phases) and key player information. Each report was categorized into these themes, and the time spent on each was examined for differences through one-way analyses of variance (ANOVA). The time spent on different domain-specific themes was also analysed and would also be used to further analyse the reports for the next part of this paper.

Results

As is visible in the distribution of time spent on different data sources in Figure 1, there is a clear dominance of qualitative video data over the two quantitative data types of event and position data in all reports. Table 1 also shows that qualitative video data occupies an average of 217 seconds and 35% of total time as against quantitative data (both event and position data combined), taking up only approximately 100 seconds and 17.5% of total time, less than half the time spent on qualitative data. The same is also supported with an independent-samples *t*-test with the opponent reports as the independent variable and the groups created based on the use of qualitative (video data, coded as '0') and quantitative (event + position data, coded as '1') data as the dependent variable with mean time spent on qualitative data being almost 2 minutes more across all the reports generated by the analysts ($t(18) = 3.922, p < 0.001$).

Another potentially insightful breakdown pertains to the content of the video reports, which have been broken down into contextually relevant content categories derived after transcribing and thematizing each report. Four major themes emerged: general information (line-ups, substitutes, formation, etc.), key player details, tactical behaviour of opponent and recommendations for own team. Each video was then broken down into these categories to arrive at the relative distribution of these themes in all the reports, which is visible in Figure 2.

Figure 2 shows that there is a diverse spread of the themes in each of the opponent reports, but there is a clear dominance of the themes 'opponent tactical behaviour' and 'recommendations for the own team'. The average time spent on opponent tactical behaviour is the highest (mean = 286.42 seconds), followed by recommendations for own team (mean = 139 seconds) and general information (mean = 76.5 seconds), and the least time

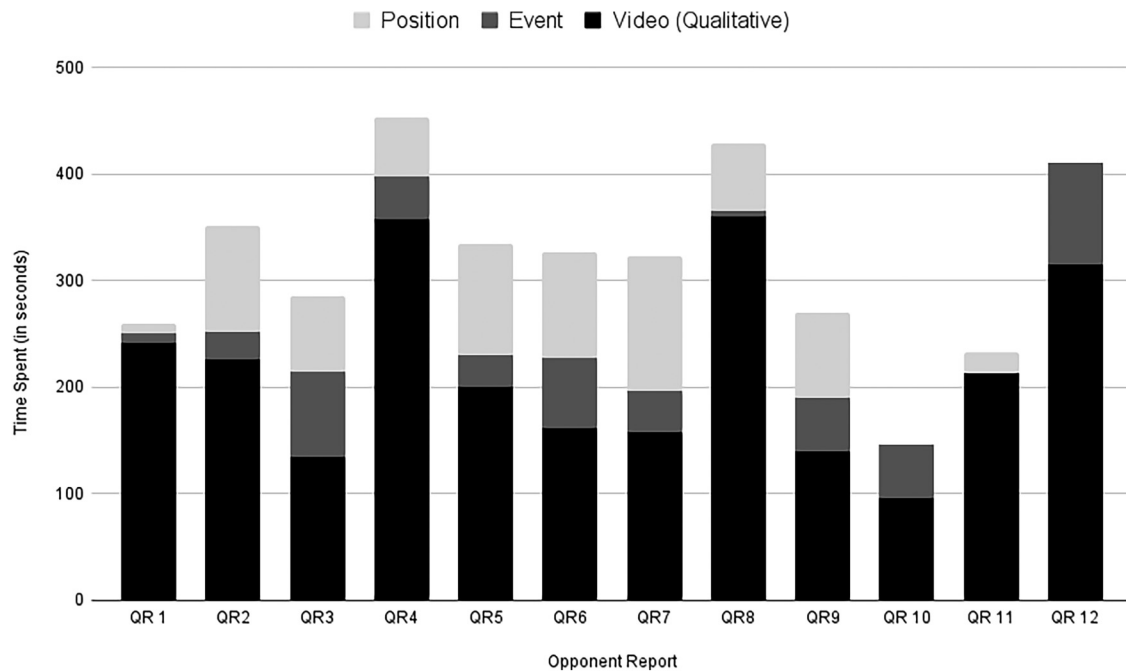


Figure 1. Time-wise (in seconds) breakdown of the mention/use of video, event and position data in each opponent report (QR1 to QR12).

Table 1. Absolute (in seconds) and percentage time spent on different data sources for each of the opponent reports coded QR1 to QR12.

Report name	Video data (in seconds)	Video data %	Event data (in seconds)	Event data %	Position data (in seconds)	Position data %	Total duration (in seconds)
QR1	242	47.17	9	1.75	9	1.75	513
QR2	226	35.82	26	4.12	99	15.69	631
QR3	135	33.58	80	19.9	70	17.41	402
QR4	358	46.19	40	5.16	55	7.1	775
QR5	200	33.11	30	4.97	104	17.22	604
QR6	162	27.09	65	10.87	100	16.72	598
QR7	158	27.34	38	6.57	127	21.97	578
QR8	360	58.25	5	0.81	64	10.36	618
QR9	140	26.72	50	9.54	80	15.27	524
QR10	96	17.55	50	9.14	0	0	547
QR11	213	36.92	0	0	20	3.47	577
QR12	315	37.15	95	11.2	0	0	848
Mean	217	35.57	40.67	7.00	60.67	10.58	601.25

spent was on key player information (mean = 44.2 seconds). The difference in time spent between groups was also reflected in one-way ANOVA with the different themes as the independent variable and the time spent on each theme as the dependent variable ($F(3,44) = 26.25$, $p < 0.001$), with all reports spending considerably higher time on team recommendations (139 ± 85.822 seconds) and opponent tactical behaviour (286.42 ± 84.142 seconds) as compared to general information (76.58 ± 55.856 seconds) and key player information (44.17 ± 57.509 seconds).

Discussion

The aim of this study was to first describe the relative usage of qualitative (video data) and quantitative (event and position data) data sources that formulate an opponent report. The frequency of time spent on either data source clearly suggests a dominance of qualitative video data that dominates the other data types, while t -test results show a clear and statistically significant difference ($p < 0.01$) in all the reports. This resembles the findings of recent work by Herold et al.¹ that noted a lower-than-expected

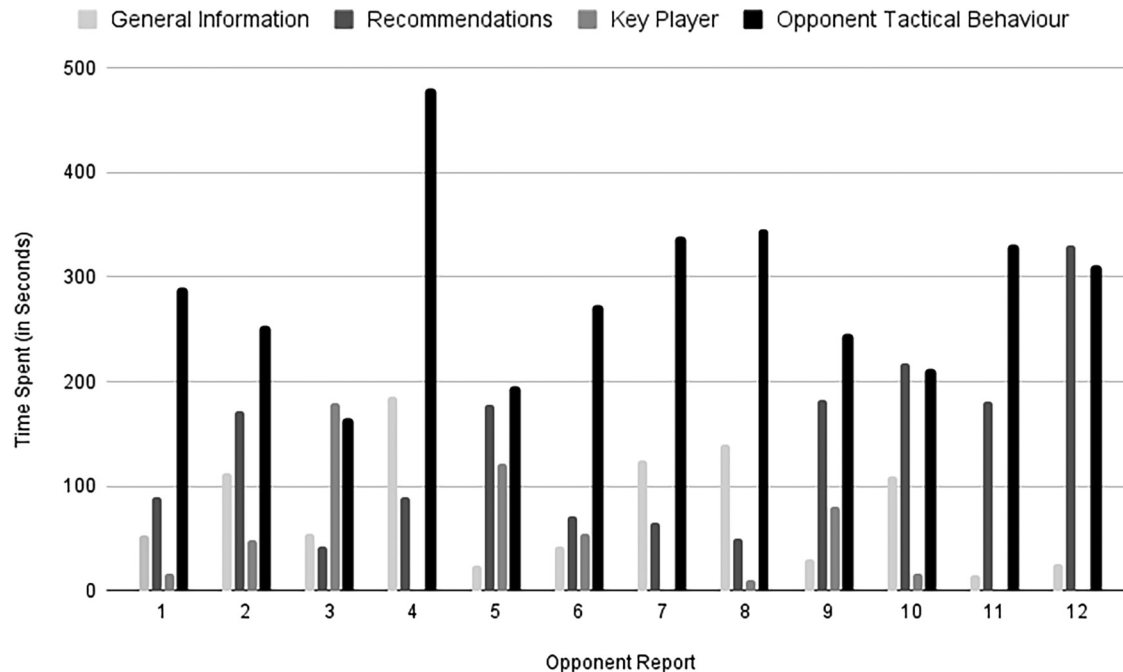


Figure 2. Time (in seconds) spent by each opponent report on themes: general information, team recommendations, key player information and opponent tactical behaviour.

implementation and perceived value of quantitative positional data when studying the use of new KPIs in the measurement of tactical performance in soccer. This dominance of qualitative data may also be driven by the preferences of coaches that prefer to use subjective scouting analysis over other data sources,⁶ while also being potentially affected by the familiarity of analysts with different data sources. Moreover, the time spent on different domain-specific themes also shows that not only is there a difference in the type of data used (qualitative vs quantitative), but there is also a difference in what the data is used to describe – some reports (QR7, QR11 and QR12) spent no time describing key player information, while another report (QR3) spent the maximum time discussing this theme as against any other theme in their report.

While these differences are not critical in determining the best and worst reports, the findings still show that there are individual differences in the choice of including different data sources, and subsequently, there is no consensus in how the data is used either. This corroborates previous work that suggests that most data must be reviewed by coaches before being included in feedback, which is why reports may be affected by subjective coach preferences and pre-determined protocols as well as knowledge and familiarity with different types of data.^{5,6} Thus, while there may be a clear increase in the volume of data types as well as the sophistication of personnel involved,^{1,17} qualitative video data still dominates opponent performance analysis, once again pointing to the gap in knowledge transfer between research and practice, while affirming the

human aspect of performance analysis.²³ Analysts may also often possess the knowledge of different data sources but may determine the content of a report keeping the coach in mind, as well as the players to whom the feedback must be delivered. With this in mind, it may be highly insightful to see how different coaches interpret the opponent reports to provide a comprehensive picture of not only the adoption of different data sources by analysts but also the interpretation of the same by coaches.

Study 2: Evaluating opponent reports by coaches in terms of match preparation

Study 1 has shown that despite providing analysts with the exact same dataset to develop an opposition report, the final product may not exhibit any uniformity in the volume of data sources used. While this is an important finding, it holds greater value in light of the interpretation of the opponent report that would be made by coaches. Thus, while the content of the reports exhibits differences, the final part of this paper asks what level the objectivity of opposition analysis is maintained when the same analysis is evaluated by different coaches. By asking expert coaches to evaluate the opposition analysis created by professional club analysts who were all given the same dataset, we examine the level of agreement amongst coaches regarding the quality of a given report. Overall grades awarded by coaches are used to gauge this level of agreement, while sub-grades awarded to each report on criteria

of comprehensibility, relevance, time adequacy and information density are used to analyse the importance of different components within each report according to different coaches. If there is no preference of style, nature and content, there would be a high agreement amongst coaches' evaluations and vice versa.

Method

In order to qualify for further coach evaluation, the report had to use at least two of the three different sources of data (video, event and position) within the opposition report, it had to be in German, and it could not exceed 15 minutes of time. Based on these criteria, the present work comprises 12 ($n=12$) recorded opponent reports. On average, the presentation time of the sample was 10:05 minutes (mean = 10:05, min = 06:44 and max = 14:08). The videos were analysed for the amount of time spent on each data type and the descriptions of what the data was used for (including the most commonly preferred KPIs within each data type), followed by a content breakdown of the videos in terms of general information, opponent tactical behaviour, general recommendations and key player information. Independent-samples *t*-tests were performed to investigate whether there were any grade-related, significant differences in the use of different data types, as well as the time-specific emphasis on certain themes within reports with ANOVA. The choice of what metrics to include was based on the analyst, but given that the dataset described the same team, the objectivity of the data could be expressed in terms of similar reports, the inclusion of certain KPIs and the guidance of recommendations. Owing to the literature regarding the complexity of coaching practices and procedural knowledge regarding the working processes of performance analysis,¹⁵ one can expect that the content and form of opposition analysis are highly dependent on the coach-analyst relationship and the overall coaching philosophy. Thus, we can hypothesize a lack of consensus in what is considered good or bad in opposition analysis. This research may also help create a consensus regarding what is considered vital (and not-so-vital) by different coaches, building towards a gold standard of opposition analysis.

Sample and data collection. Thirty-six football coaches who train teams involved in different professional leagues within Germany were included in this part of the study. These coaches (average age = 30.43) had an average of 7 years of coaching experience, with a B licence or higher, in different German leagues. They were asked to rate the opponent report that was created by analysts in Study 1. Each participant was randomly assigned any three of the existing 12 reports to grade. A random classifier was used to assign reports to each participant, such that each of the 12 videos was at least graded nine times. With an online

questionnaire hosted on SoSci Survey (soscisurvey.de), the grades were compiled and aggregated along with coach demographics.

Procedure. In order to maintain the intuitiveness of responses, participants were asked to grant overall grades for each opponent report based on the German school grading system that they were mostly used to. Ranging from 6 to 1, with 1 being the best and 6 being the worst grade, the participants were asked to view each report from start to finish and provide a single overall grade as a general assessment. Besides the overall grade, the opponent reports also had to be rated for four sub-components: relevance of the content, comprehensiveness of the overall report, time adequacy of the report and information density of the report. These sub-components were collected with a Likert-type rating scale (1 = *poor* and 5 = *excellent*).

Data analyses. The obtained grades for each of the reports were compared in terms of range and accuracy. In other words, the level of agreement amongst different overall grades for the same video was analysed. Given that all the aforementioned data was based on qualitative ratings, the descriptive analysis of mean and standard deviation (SD) would provide a basic understanding of levels of agreement, alongside performing an inter-rater reliability test for the videos through a single-measures intra-class correlation (ICC) between all the awarded grades for each report, which may help us further examine the clustering of scores. Additionally, the best- and worst-rated videos would also be compared for levels of agreement. Furthermore, the sub-grades of relevance, comprehensibility, time length and information density were also used in an independent-samples *t*-test to identify which of the sub-grades most contributed to the overall grade. This was done by comparing high- and low-graded report groups (with mean and standard deviation) that were used as the two groups for the *t*-test. Finally, the open-ended question of ascribing important components to the graded videos was analysed through a thematic content breakdown that identified major themes, the frequency of their usage and consequently the diversity of topics that were mentioned.

Results

Table 2 shows the breakdown of different overall grades awarded to each of the reports, along with their mean (and SD) scores. Each row of reports consists of the number of coaches that awarded the report every grade from 1 to 6. As the overall grades awarded to each report are based on the German school grading system, the lower the average grade, the better is the rating. The best-rated videos are QR2 (six coaches awarded it a 2, and three coaches awarded it a 3; mean grade = 2.33, SD = 0.5) and QR11 (mean = 2.33, SD = 0.71), followed by

Table 2. Number of times each overall grade (1–6, as columns) was given to reports (QR1 to QR12) by different coaches, along with the mean (and standard deviation (SD)) grade for each report.

Report	Overall grade						Total number of grades given	Mean	SD
	1	2	3	4	5	6			
QR1		3	4	2			9	2.89	0.78
QR2		6	3				9	2.33	0.50
QR3		4	4	2	1		11	3.00	1.00
QR4	2	4	3		2		11	2.64	1.36
QR5		5	4	1	1		11	2.82	0.98
QR6	1	2	4		2		9	3.00	1.32
QR7			4	2	3	1	10	4.10	1.10
QR8			3	5	1		9	3.78	0.67
QR9		1	7		1		9	3.11	0.78
QR10		3	3	1		2	9	3.44	1.59
QR11	1	4	4				9	2.33	0.71
QR12		2	4	2	2	1	11	3.64	1.29

QR4 (mean = 3, SD = 1), while the worst-graded videos are QR7 (mean = 4.1, SD = 1.1) and QR8 (mean = 3.78, SD = 0.67), followed by QR12 (mean = 3.64, SD = 1.29). QR12 is also a good example of looking closer into these descriptive statistics to inform us about levels of agreement of grade, with different coaches awarding that report every grade from 2 to 6. This type of breakdown of different grades for each report is depicted in Figure 3, which shows multiple reports getting awarded 4–5 different grades. The report with the least diversity of grades is also the highest-rated (QR2), and similarly, low diversity is observed in the other highly graded report QR11 and the worst-graded report QR8. Seven out of the 12 total reports show 4 or more different grades awarded to the reports. A single-measures intra-class correlation (Table 3) between the grades awarded by different coaches for each report also showed a weak (and statistically significant) correlation ($r = 0.137$; $p = 0.011$), further clarifying the lack of agreement between coaches in the grading of these reports. It can also be argued that the precision seen in the grading of the best and worst overall reports is higher than the rest of the reports, as is visible in the SD of QR2 (SD = 0.50), QR11 (SD = 0.71) and QR8 (SD = 0.67) as well as the respective bars (QR2, QR11 and QR8) in Figure 3.

Figure 4 shows the mean sub-grade component of relevant aspects, comprehensibility, report being too long, and report being too dense. Better-rated videos like QR2 and QR11 show high scores on relevance (QR2 mean = 3.56, SD = 0.53; QR11 mean = 3.44, SD = 1.24) and comprehensibility (QR2 mean = 3.89, SD = 0.6; QR11 mean = 3.33, SD = 1.41) and lower scores on the reports being too long (QR2 mean = 3.11, SD = 1.17; QR11 mean = 2.78, SD = 0.97) and dense (QR2 mean = 2.89, SD = 1.05; QR11 mean = 2.78, SD = 1.2), whereas the badly

rated reports like QR7 and QR8 show similar scores in all sub-components, indicating comparatively lower levels of relevance (QR7 mean = 2.8, SD = 1.23; QR8 mean = 2.8, SD = 0.53) and comprehensibility (QR2 mean = 2.8, SD = 1.14; QR8 mean = 2.8, SD = 0.87) and the reports being too long (QR7 mean = 3, SD = 1.15; QR8 mean = 3, SD = 1.2) and dense (QR7 mean = 2.9, SD = 1.1; QR8 mean = 2.9, SD = 1.59). To gain a further understanding of how these sub-components affect the grading of the best and worst videos, an independent-samples *t*-test was performed for groups of high-graded (overall grade 2 or lower) and low-graded reports (overall grade 4 or higher) to understand which sub-component distinguishes these videos the most. Results from this test show that a statistically significant difference between the high-graded and low-graded reports exists in sub-components of relevance ($t(68) = 3.563$, $p = 0.001$) and comprehensibility ($t(68) = 5.41$, $p < 0.001$) but not in the grades awarded to the reports for their length adequacy ($t(68) = -1.765$, $p = 0.082$) or information density ($t(68) = -1.573$, $p = 0.12$).

Discussion

The aim of this study was to examine the level of agreement in the evaluations of the same opponent report made by different coaches. Twelve different opponent reports were each evaluated by at least nine coaches with overall grades as well as sub-grades, after which a single-measures intra-class correlation test (Table 3) was used to establish the level of agreement in the grading of the same report. The findings suggest a clear and significant ($p = 0.011$) lack of consensus in the grades provided by different coaches for the same report. This goes to show that the level of agreement between coaches regarding the quality of an opponent report is also highly subjective. The same is also seen in Figure 3, where seven out of 12 reports have been given 4 or more different grades. It seems pretty apparent that each coach has a preference of how a report must be formulated and what the report must emphasize, which is associated with the coaching philosophy as well as the analyst-coach relationship within the larger social structure of different clubs.^{5,6,8} This also makes sense from a value co-creation perspective that highlights the importance of relationships and interactions of all actors within a system, yielding value as a result of exchange and collaboration.¹¹ Therefore, it would make sense that coaches from different clubs have different opinions and preferences in how they evaluate a good (or bad) report. As for the mean sub-grades awarded to each report, as is visible in Figure 4, the best- (QR2, QR11) and worst-graded (QR7, QR8) videos show differences in components of relevance and comprehensibility. Independent-samples *t*-tests comparing groups of the best- and worst-graded videos also show a statistically significant difference in mean scores of relevance ($t(68) = 3.563$, $p = 0.001$) and comprehensibility ($t(68) = 5.41$, $p <$

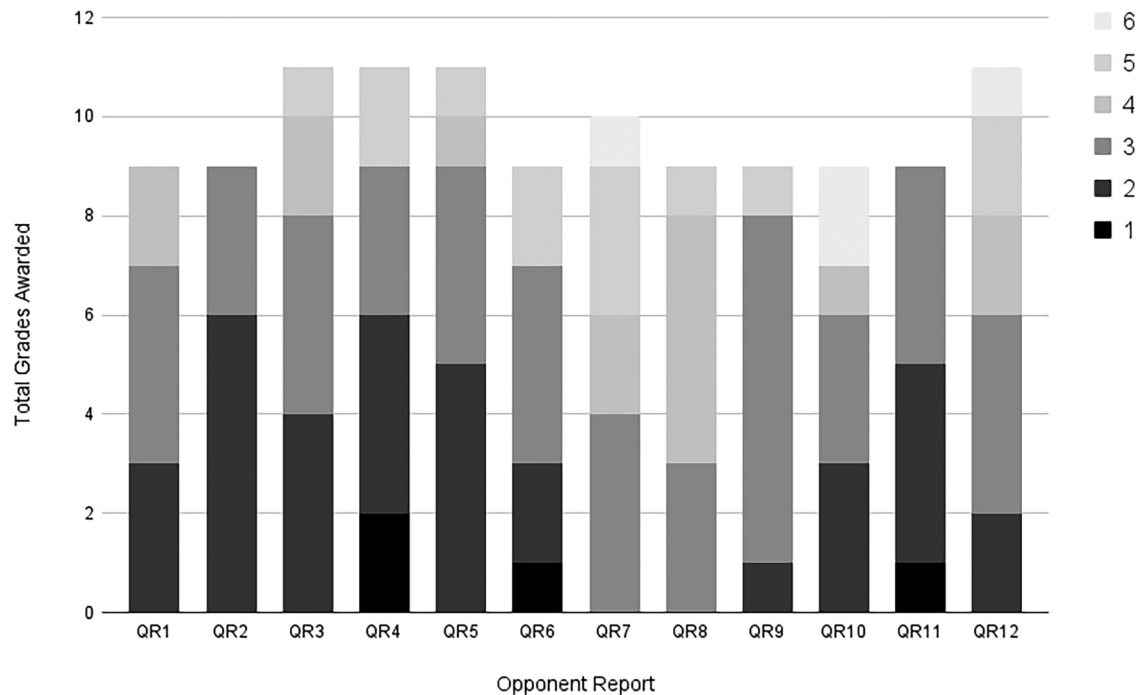


Figure 3. Breakdown of different overall grades awarded to each opponent report (QR1 to QR12) by every coach.

Table 3. Intra-class correlation results for different overall grades awarded to each opponent report (QR1 to QR12). As this study treats each grade individually and not an average of multiple grades, the single-measures output must be considered.

		95% confidence interval		F-test with true value 0			
	Intra-class correlation ^b	Lower bound	Upper bound	Value	df1	df2	Sig.
Single measures	0.137^a	0.015	0.408	2.432	11	88	0.011
Average measures	0.589 ^c	0.119	0.861	2.432	11	88	0.011

Two-way mixed-effects model where people effects are random and measures effects are fixed.

^aThe estimator is the same whether the interaction effect is present or not.

^bType C intra-class correlation coefficients using a consistency definition. The between-measures variance is excluded from the denominator variance.

^cThis estimate is computed assuming the interaction effect is absent because it is not estimable otherwise.

0.001) and not in time adequacy and information density. This would imply that, according to the sub-grades awarded by coaches, the best- and worst-graded videos differ in terms of comprehensibility and relevance and not so much in terms of their length or density of information. These results yet again generally suggest that possessing the knowledge of the coaching philosophy is naturally imperative in adequate and efficient feedback from staff to players,^{5,13,14} while illuminating the individuality of coaching preferences when it comes to creating meaningful information and value.¹¹

General discussion

The aim of the study was to examine the use of different data sources by analysts in creating an opponent report and the evaluation of these by coaches to gauge the process of developing meaningful information for football

players. The results from both inquiries show a clear lack of agreement in the use of different data sources, as well as the coach grades awarded to them, indicating the presence of an inherent subjectivity that is characteristic of any coach's philosophy or principles. It also shows that there may be numerous individual preferences in terms of determining what metrics indicate successful or desirable performance, which must be discussed and negotiated with all members of the coaching staff.^{1,24,25} This process of exchange sheds light on the importance of considering the development of meaningful information as an interactive, co-creative process involving multiple stakeholders while also considering the context in which this develops.^{10,11} The effect of social processes and social environments has been explored more recently in the setting of a football academy,⁸ where the effect of social hierarchies, social capital and symbolic capital has been studied with respect to the development of coaching staff's (assistant

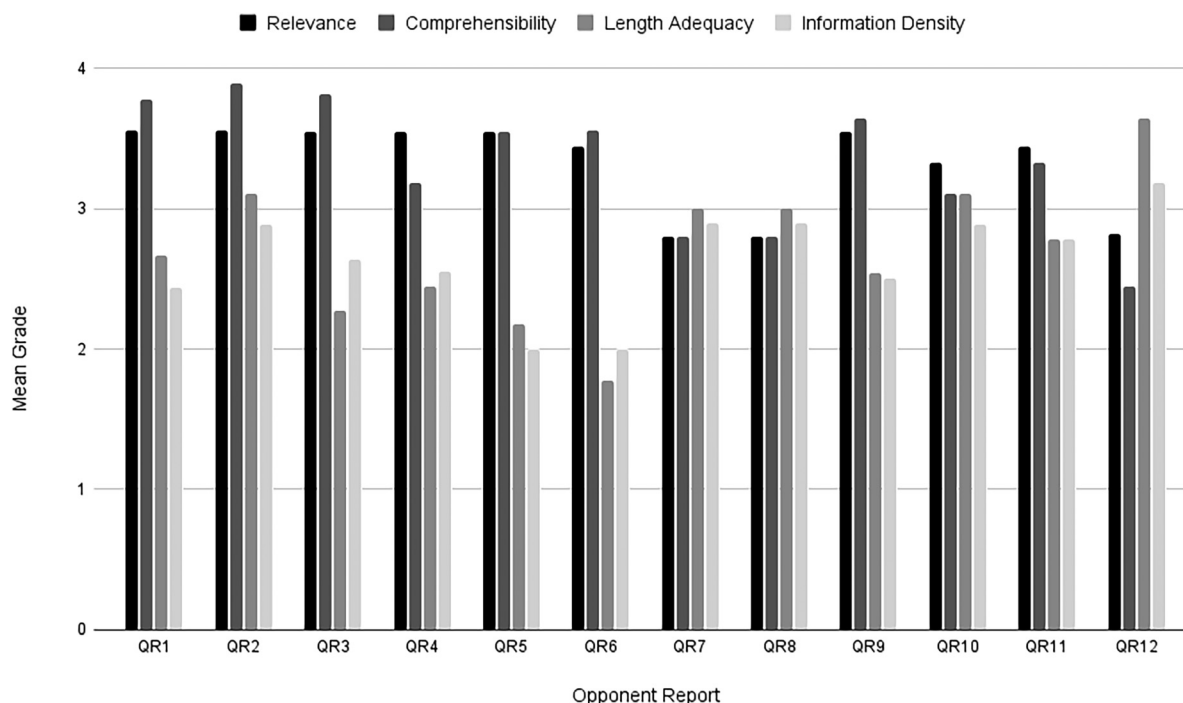


Figure 4. Breakdown of mean sub-grades awarded to each opponent report (QR1 to QR12) on criteria of relevance, comprehensibility, length adequacy and information density of each report.

coaches, analysts and scouts) education, ideological belief systems and ‘philosophies’.⁸ Given that most of the analysts in our study were involved in some capacity with professional clubs, these individual differences can be better understood, keeping in mind the potential differences in the philosophy of their respective clubs. Consequently, this may also mean that the general belief of greater objectivity accrued to the use of quantitative data may not be accurate due to multiple reasons like a lack of transfer from research to practice,²³ underlying coach–analyst preferences,⁵ coaching philosophies⁸ and preferred modes of feedback to players.^{5,6,13}

Recent work in understanding the use of different KPIs has already revealed that despite having complex KPIs at their disposal, practitioners often prefer using rather basic or simple quantitative performance measures (like shot on goal within event data).¹ This once again raises the question of transfer of knowledge while elucidating the human component that characterizes opponent match analysis at different levels: adoption of different metrics as sources of description, interpretation of associated characteristics of an opponent and preferences of what formulates good and bad opponent reports. This involves analysts at junior and senior levels, coaches and their assistants as well as the players who ultimately receive and attempt to integrate their feedback. With the expectation that the future yields even more complex KPIs, analysts will continue to serve as the human interface within technology.⁹ This sheds light on what has previously been called the black box of the

performance analytical process⁷ and corroborates the need for comprehensive frameworks that clarify the role of performance analysts as well as the variety of skills that may be required of them in the future.^{9,10} This study furthers previous findings that show the variability associated with the adoption of different KPIs and goes to show that high volumes of data may not necessarily translate to objective conclusions, despite originating from novel and complex data sources.^{26,27} In this regard, it may be valuable to consider analysts as service providers that need to customize and individualize their contributions to the decision-making process based on the preferences and needs of involved stakeholders. Future research should build in this direction of deconstructing the complexity of this process, such that the fundamental requirements can be derived and possibly improved upon. This would not only help decomplexify the process but also serve as a blueprint of value creation for young and upcoming analysts.¹⁰

Future directions

The next steps in this direction should shed light on the specifics of coach–analyst interactions and relationships, the processual flow that characterizes the relationship and how the athlete fits into the framework as new methods like player video sessions become a mainstay in building up to every game. A deeper look at the content of the best-rated and worst-rated reports may also provide insights into certain themes or domains that relate to higher grades.

This could potentially help define critical elements of opponent analysis, though the method must be supported and clarified with a larger dataset of coaches. Some initial indicators for future studies can be derived from the current data, such as the finding that the coaches in this sample most value the relevance and comprehensibility of the reports. Such qualities of the reports can be combined with specific themes to develop an application-oriented opponent report design in a bid to provide greater structure to this aspect of applied performance analysis. This could be of great value to develop pathways of integrating research and application, such that the gulf between the presence of different data types and the use of these on-field can be further synchronized.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship and/or publication of this article.

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship and/or publication of this article: This project was supported by the German Federal Ministry of Education and Research (BMBF – Bundesministerium für Bildung und Forschung) under Grant 01IS20021A for the last author.

ORCID iDs

Saumya Mehta  <https://orcid.org/0000-0001-9187-7729>

Philip Furley  <https://orcid.org/0000-0002-1230-2220>

References

1. Herold M, Kempe M, Bauer P, et al. Attacking key performance indicators in soccer: current practice and perceptions from the elite to youth academy level. *J Sport Sci Med* 2021; 20: 158–169. Available from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7919358/>.
2. McKenna M, Cowan DT, Stevenson D, et al. Neophyte experiences of football (soccer) match analysis: a multiple case study approach. *Res Sports Med* 2018; 26: 306–322.
3. Cushion C. Modelling the complexity of the coaching process. *Int J Sports Sci Coach* 2007; 2: 395–401.
4. Jones RL, Armour KM and Potrac P. Understanding the coaching process: a framework for social analysis. *Quest* 2002; 54: 34–48.
5. Wright C, Atkins S and Jones B. An analysis of elite coaches' engagement with performance analysis services (match, notational analysis and technique analysis). *Int J Perform Anal Sport* 2012; 12: 436–451.
6. Nelson LJ and Groom R. The analysis of athletic performance: some practical and philosophical considerations. *Sport Educ Soc* 2011; 17: 687–701.
7. Bampouras MT, Cronin C and Miller KP. Performance analytic processes in elite sport practice: an exploratory investigation of the perspectives of a sport scientist, coach and athlete. *Int J Perform Anal Sport* 2012; 12: 468–483.
8. Hall J, Cope E, Townsend RC, et al. Investigating the alignment between coaches' ideological beliefs and academy philosophy in professional youth football. *Sport Educ Soc* 2020; 27: 377–392.
9. Robertson PS. Man & machine: adaptive tools for the contemporary performance analyst. *J Sport Sci* 2020; 38: 2118–2126.
10. Martin D, O Donoghue PG, Bradley J, et al. Developing a framework for professional practice in applied performance analysis. *Int J Perf Anal Sport* 2021; 21: 845–888.
11. Vargo SL, Akaka MA and Vaughan CM. Conceptualizing value: a service-ecosystem view. *J Creating Value* 2017; 3: 117–124.
12. Wright C, Atkins S, Jones B, et al. The role of performance analysts within the coaching process: performance analysts survey 'the role of performance analysts in elite football club settings'. *Int J Perf Anal Sport* 2013; 13: 240–261.
13. Soccer analytics: the money ball of football. *Sports analytic conference* <http://www.youtube.com/watch?v=YaTALjKvAQ> (2013, accessed 24 June 2023).
14. Groom R and Cushion C. Coaches perceptions of the use of video analysis: a case study. *Insight* 2004; 3: 56–59.
15. Groom R, Cushion C, Nelson L, et al. The delivery of video-based performance analysis by England youth soccer coaches: towards a grounded theory. *J Appl Sport Psychol* 2011; 23: 16–32.
16. Fujimura A and Sugihara K. Geometric analysis and quantitative evaluation of sport teamwork. *Syst Comput Jpn* 2011; 36: 49–58.
17. Rein R and Memmert D. Big data and tactical analysis in elite soccer: future challenges and opportunities for sports science. *SpringerPlus* 2016; 5: 1–13.
18. Smith B and McGannon KR. Developing rigor in qualitative research: problems and opportunities within sport and exercise psychology. *Int Rev Sport Exerc Psychol* 2018; 11: 101–121.
19. Tracy SJ. Qualitative quality: eight "big-tent" criteria for excellent qualitative research. *Qual Inq* 2010; 16: 837–851.
20. Marois R and Ivanoff J. Capacity limits of information processing in the brain. *Trends Cogn Sci* 2005; 9: 296–305.
21. Silver N. *The signal and the noise: the art and science of prediction*. London: Penguin group, 2015.
22. Robinson OC. Sampling in interview-based qualitative research: a theoretical and practical guide. *Qual Res Psychol* 2014; 11: 25–41.
23. Mackenzie R and Cushion C. Performance analysis in football: a critical review and implications for future research. *J Sport Sci* 2013; 31: 639–676.
24. Memmert D (Ed.). *Match analysis*. Abingdon: Routledge, 2021.
25. Memmert D and Raabe D. *Data analytics in football. Positional data collection, modelling and analysis*. Abingdon: Routledge, 2018.
26. Low B, Coutinho D, Gonçalves B, et al. A systematic review of collective tactical behaviours in football using positional data. *Sports Med* 2019; 50: 343–385.
27. Memmert D, Lemmink KAPM and Sampaio J. Current approaches to tactical performance analyses in soccer using position data. *Sports Med* 2017; 47: 1–10.